

Kaif Hazarika

Atmospheric & Space Science Researcher

Profile

Physics researcher specializing in **upper-atmosphere and space physics**, with undergraduate and postgraduate research into thermospheric wind response to geomagnetic storms and equatorial $E \times B$ drift dynamics using satellite (C/NOFS) data. Extends this into applied climate research – a state-government climate-resilience fellowship and a satellite-data case study on climate change's effects on rural livelihoods – alongside two filed patents in emissions estimation and information systems, and hands-on library and information systems experience. Proficient in **Python**, **Fortran**, **C++**, and **Java**. Seeking a research, analysis, or field role in atmospheric or climate science.

Research Interests

Atmospheric Physics · Space Weather · Ionosphere–Thermosphere Coupling · Geomagnetic Storms · Climate Change & Resilience · Satellite Remote Sensing · Climate Data Analysis · Data Science for Earth Systems · Scientific Computing · Environmental Monitoring

Education

- 2018–2023 **Integrated MSc in Physics**, Dibrugarh University, Dibrugarh, Assam, Percentage: 78%
Core coursework and research in space, atmospheric, and plasma physics; final-year research on ionospheric and thermospheric dynamics.
- 2018–2021 **BSc in Physics**, Dibrugarh University, Dibrugarh, Assam, Percentage: 79.8%
- 2017 **Higher Secondary (10+2), Science**, Jawahar Navodaya Vidyalaya (JNV) Dhing, Nagaon, Assam, Percentage: 74%
- 2015 **HSLC (Class 10)**, Jawahar Navodaya Vidyalaya (JNV) Dhing, Nagaon, Assam, CGPA: 8.6

Research Projects

- 2024 **The Response of Thermosphere Meridional Wind to the Geomagnetic Storm of November 2021**, Department of Physics, Dibrugarh University, Supervisor: Dr. B. R. Kalita, Associate Professor, Dept. of Physics, Dibrugarh University Integrated MSc dissertation investigating thermospheric meridional wind response during a major geomagnetic storm, using upper-atmosphere wind and storm-index data.
- 2025 **Investigating Climate Change and Its Effects on Rural Livelihoods: A Case Study of Bhojo Village**, PGDRD Research Project (MRDP-205), IGNOU, Supervisor: Dr. Murchana Gogoi, Dept. of Sociology, Silapathar College Paired household-survey fieldwork with satellite-derived analysis of temperature trends, precipitation patterns, and greenhouse-gas concentrations to assess climate impacts on a rural community's livelihood.
- 2023 **Vertical $E \times B$ Drift Velocity Variations Using C/NOFS Satellite Data**
 - Climate Change: Adaptation, Mitigation, and Sustainability
 - Climate Data Handling and Prediction
- 2025–2026 **Impact of ICT on Library and Information Services: A Study of the Central Library of ADP College**, Indira Gandhi National Open University (IGNOU), Supervisor: Hrishikesh Bhuyan, Librarian, Anandaram Dhekial Phookan College, Nagaon, Assam
MLIS project investigating the impact of Information and Communication Technology (ICT) on library services and infrastructure.

Fellowship

- Mar 2023 – Feb 2024 **Chief Minister's Climate Resilient Village Fellowship**, Science, Technology & Climate Change Dept., Govt. of Assam, conducted by Assam Science Technology & Environment Council, Guwahati, Assam, Certificate awarded 28 Feb 2024
Community-level fellowship equipping recipients with practical skills in climate-resilience strategy and implementation at the village level.

Patents

- Filed 6 Sep 2024 **A System for Estimating Carbon Emission and Offsetting**, Application No. 202441065196 · Published 13 Sep 2024
A system for quantifying greenhouse gas emissions in the oil and gas Exploration & Production (E&P) sector, generating precise emission estimates and a tree-planting-based carbon-offset recommendation to support emission-reduction planning.

Filed 28 May 2026 **A Hardware-Integrated Offline User and Inventory Management System and Method Thereof**, *Application No. 202631066962 · Published 12 Jun 2026*
A standalone, network-independent system for entry-gate authentication and inventory control: decrypts scanner input and validates identity entirely in local memory, executes transactions against RAM-resident data structures, and generates its own encrypted machine-readable tokens without relying on an external database.

Additional Qualifications

- 2025–2026 **MLIS – Master of Library & Information Science**, *Indira Gandhi National Open University (IGNOU)*, Study Centre: Dibrugarh University, Coursework & project (MLIP002) complete; final certification in progress
- 2023–2025 **Bachelor of Library and Information Science**, *Indira Gandhi National Open University (IGNOU)*, 66.64% – First Division
- 2024–2025 **Post Graduate Diploma in Rural Development**, *Indira Gandhi National Open University (IGNOU)*, Overall 61.21% – First Division
- 2025 **Assamese Language Learning Course**, *Asam Sahitya Sabha Mukta Siksha Samsad*
Ajmal Foundation Centre, Hojai
- 2023–2024 **Diploma in Oil Well Drilling Technology**, *Department of Petroleum Technology, Dibrugarh University*, Dibrugarh, Assam, CGPA: 9.4
- 2022 **Post Graduate Diploma in Computer Application**, *National Board of Computer Education*, Percentage: 82.5%

Work Experience

- Jan 2024 **Winter Intern, Drilling Department**, *Oil India Limited*, Duliajan, Assam
One-month internship across drilling services – rig building, drilling equipment, instrumentation, directional drilling, operational procedures, and HSE protocols, including site visits to two active drilling rigs.

Technical Skills

- Atmospheric & Space Science Ionospheric physics, thermospheric dynamics, geomagnetic storm analysis, equatorial $E \times B$ drift, C/NOFS satellite data
- Climate & Environmental GHG emission estimation, satellite climate-data analysis, temperature & precipitation trend analysis, climate-resilience planning
- Remote Sensing & GIS NASA Giovanni, Google Earth Engine, NASA Worldview, Panoply, QGIS
- Data Science & AI Python data-science stack (NumPy, Pandas, Matplotlib), machine-learning fundamentals (scikit-learn), applied statistics, data-driven climate & environmental modeling
- Library & Information Science Koha (ILS), DSpace (institutional repository), MARC21 cataloguing, DDC classification, circulation & patron management, information retrieval systems, research methodology, technical writing
- Programming Python, Fortran, C++, Java, HTML, CSS, JavaScript
- Simulation & Tools Matlab, Halliburton Landmark WellPlan™, MRST-co2lab, Mudware, GNOME, GPS-X, STOAT, Design-Expert, Grapher, LaTeX, Excel, Photoshop, Unity

Certificates & Trainings

- Online Short Course on Planetary Science, CSSTEAP under Physical Research Laboratory (PRL), Dehradun, 2023
- Workshop on Space and Atmospheric Sciences Research Tools, Department of Physics, Dibrugarh University, 2023
- Workshop on Popularization of Plasma Physics, Department of Physics, Dibrugarh University & CPP-IPR, Guwahati, 2023
- Hands-on Training Program on Mitigating Carbon Emissions, FIPI Dibrugarh University Student Chapter, 2024
- 1st Position, Advaya Paper Presentation Contest, FIPI Dibrugarh University Student Chapter, 2023
- Certificate Course in Entrepreneurship (CCE), Srini Raju Centre for IT and the Networked Economy, Indian School of Business & CSC Academy
- 1-Day AI Tools Workshop, BE10X, 3 March 2024
- DST–PURSE “Synergies in Science: A One-Day Industry-Academia Exchange,” Dibrugarh University, 18 March 2024
- National Cadet Corps (NCC) ‘C’ Certificate, 2023